

GPU Architecture

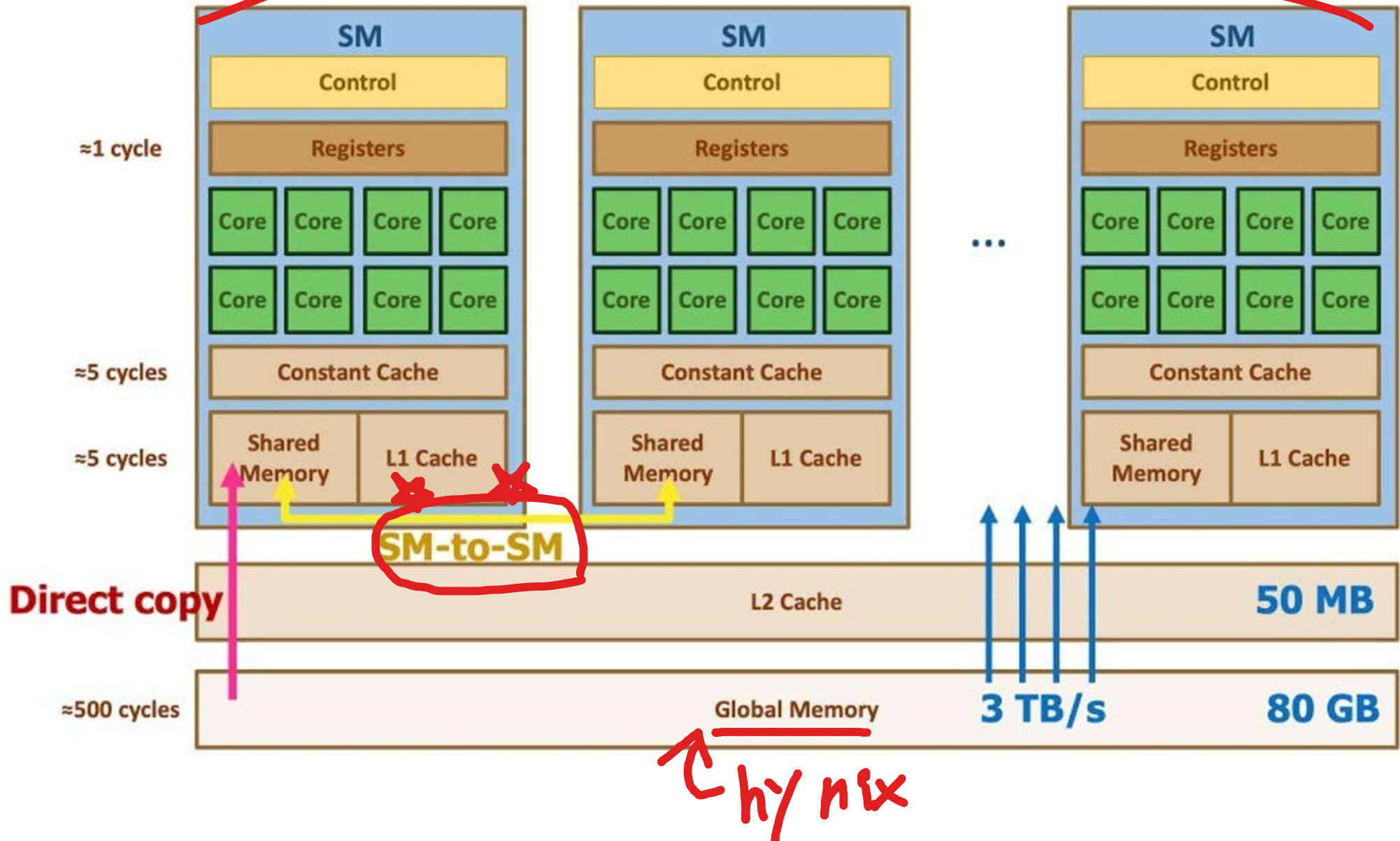
Youngjae Kim, Ph.D.

Linux 운영체제 및 응용 (GITF315-01)

Nvidia H100 GPU Architecture

같은 번에 연산

CPU와 유사함..



Overall GPU Architecture

1. Streaming Multiprocessors (SMs)

- Each SM is an independent execution unit that can run many threads in parallel.
- Inside each SM, we have several key components:
 - CUDA cores (ALUs)
 - These are the basic arithmetic units that perform integer and floating-point operations. Each SM contains tens to hundreds of CUDA cores.
 - Warp Scheduler
 - This unit selects which group of threads—called a *warp*—to execute at each cycle. It is responsible for keeping the SM busy by rapidly switching between warps.
 - Register File
 - A large on-chip storage that holds per-thread data. Since thousands of threads can be active, the register file is quite large.
 - Shared Memory (on-chip SRAM)
 - A fast, programmer-managed memory that allows threads within the same block to share data efficiently. It has much lower latency than global memory.

Overall GPU Architecture

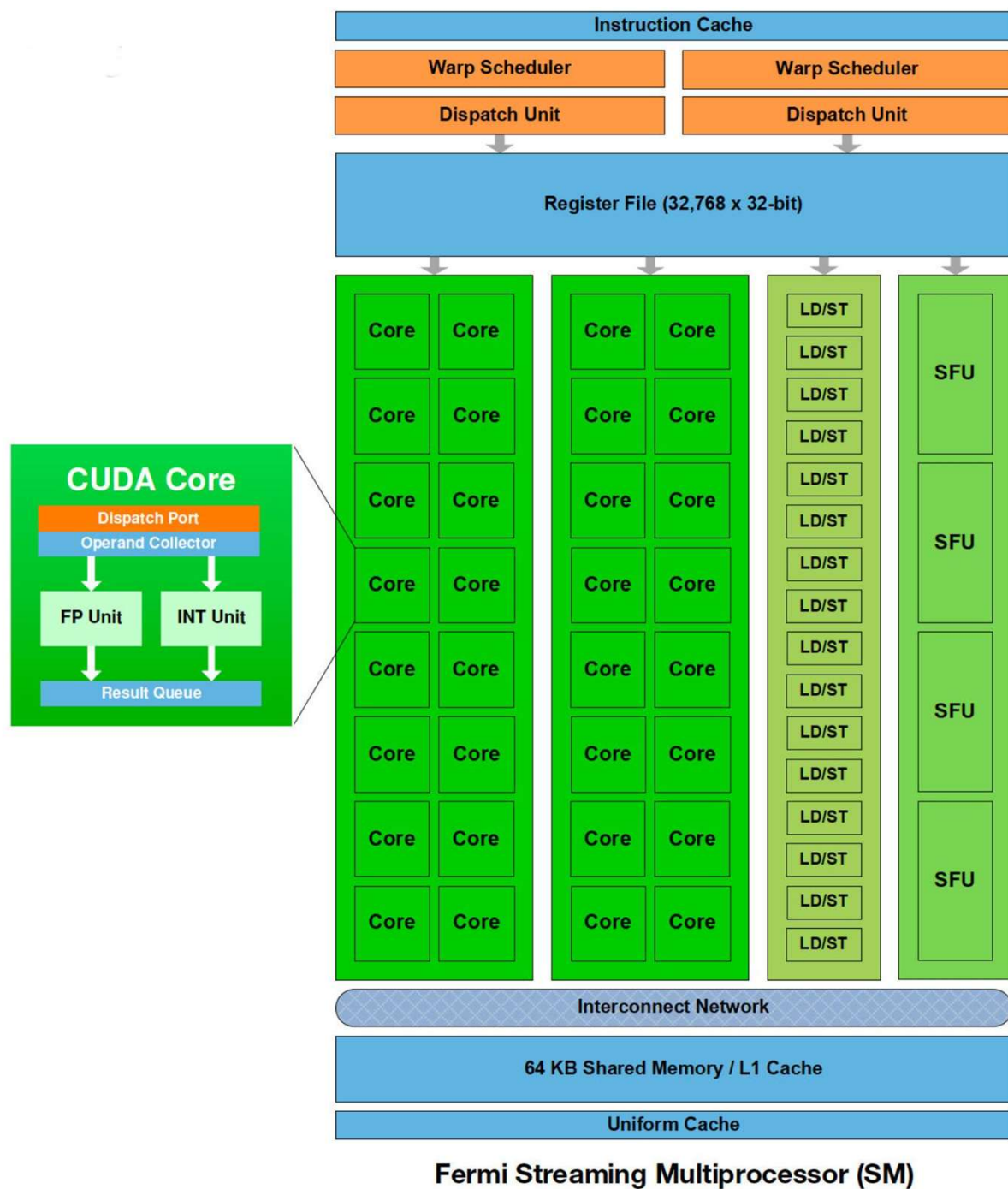
2. GPU-wide Components

- **Beyond individual SMs, the GPU also includes shared resources:**
 - **L2 Cache (shared across SMs)**

This is a global cache that sits between the SMs and main memory.
It helps reduce memory access latency and traffic to DRAM.
 - **Global Memory (HBM or GDDR)**

This is the off-chip memory of the GPU.
It provides large capacity but has much higher latency compared to on-chip memories.

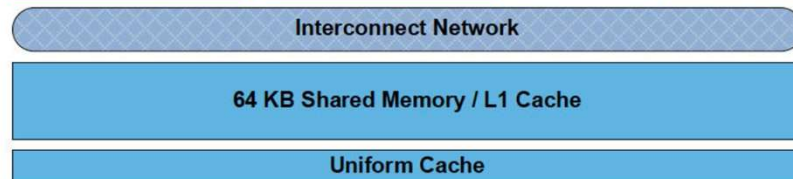
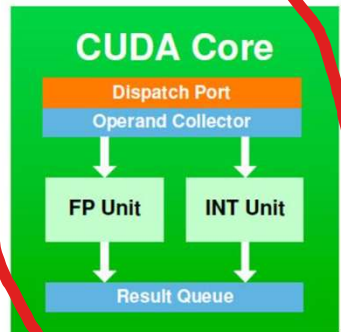
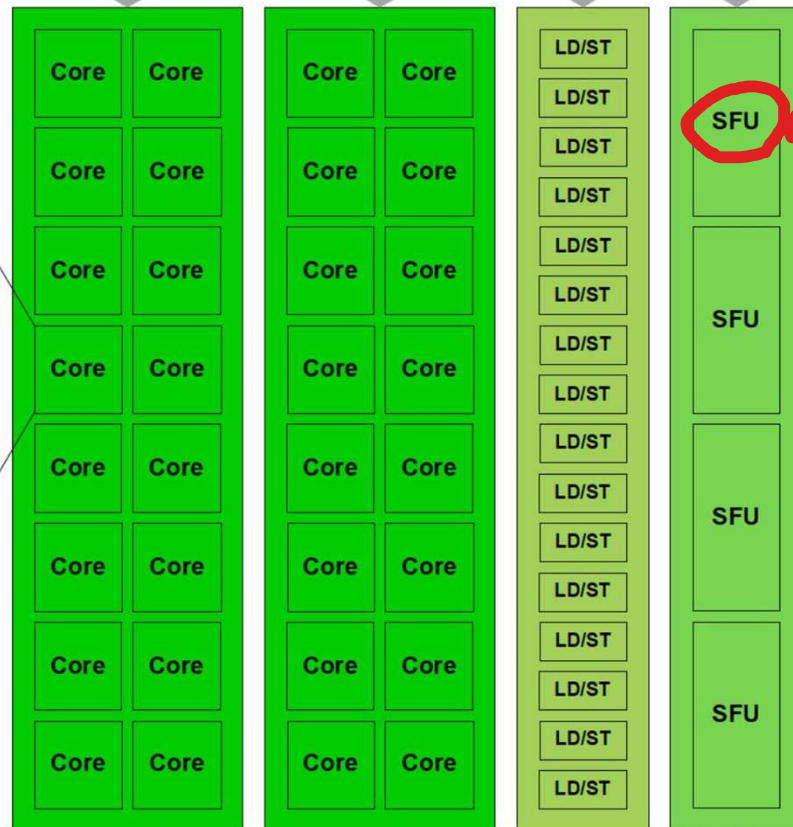
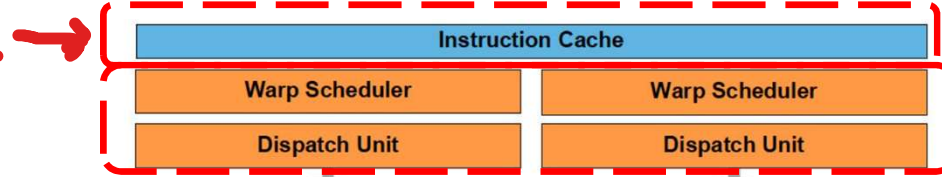
Architecture of a Fermi Streaming Multiprocessor (SM)



Fermi SM CUDA Architecture
Release date: March 2010 (GTC 2010)

Architecture of a Fermi Streaming Multiprocessor (SM)

명령어



Fermi Streaming Multiprocessor (SM)

1. Instruction Supply

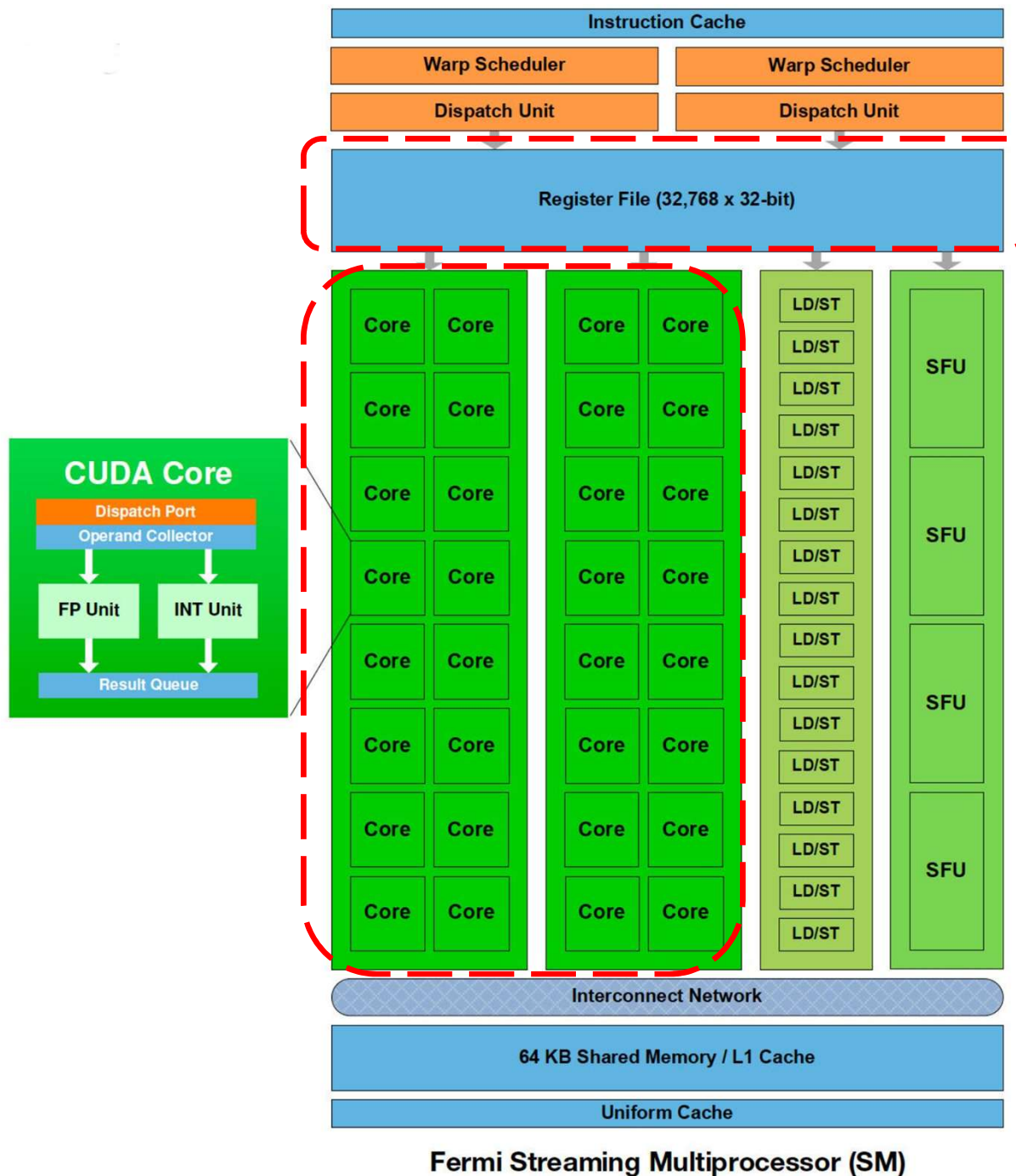
- **Instruction Cache.**
- This is where the SM fetches instructions that will be executed by threads.
It works similarly to the instruction cache in a CPU.

Special Function Unit : ALU와 같은 연산기

2. Warp Scheduling (The Brain)

- **Two Warp Schedulers and Dispatch Units.**
- A key concept in GPUs is the **warp**, which is a group of 32 threads executed together.
- The **warp scheduler** selects which warp to run at each cycle.
- The **dispatch unit** sends the selected instruction to execution units.
- Unlike CPUs, the GPU doesn't focus on a single thread.
Instead, it rapidly switches between many warps to keep the hardware busy.

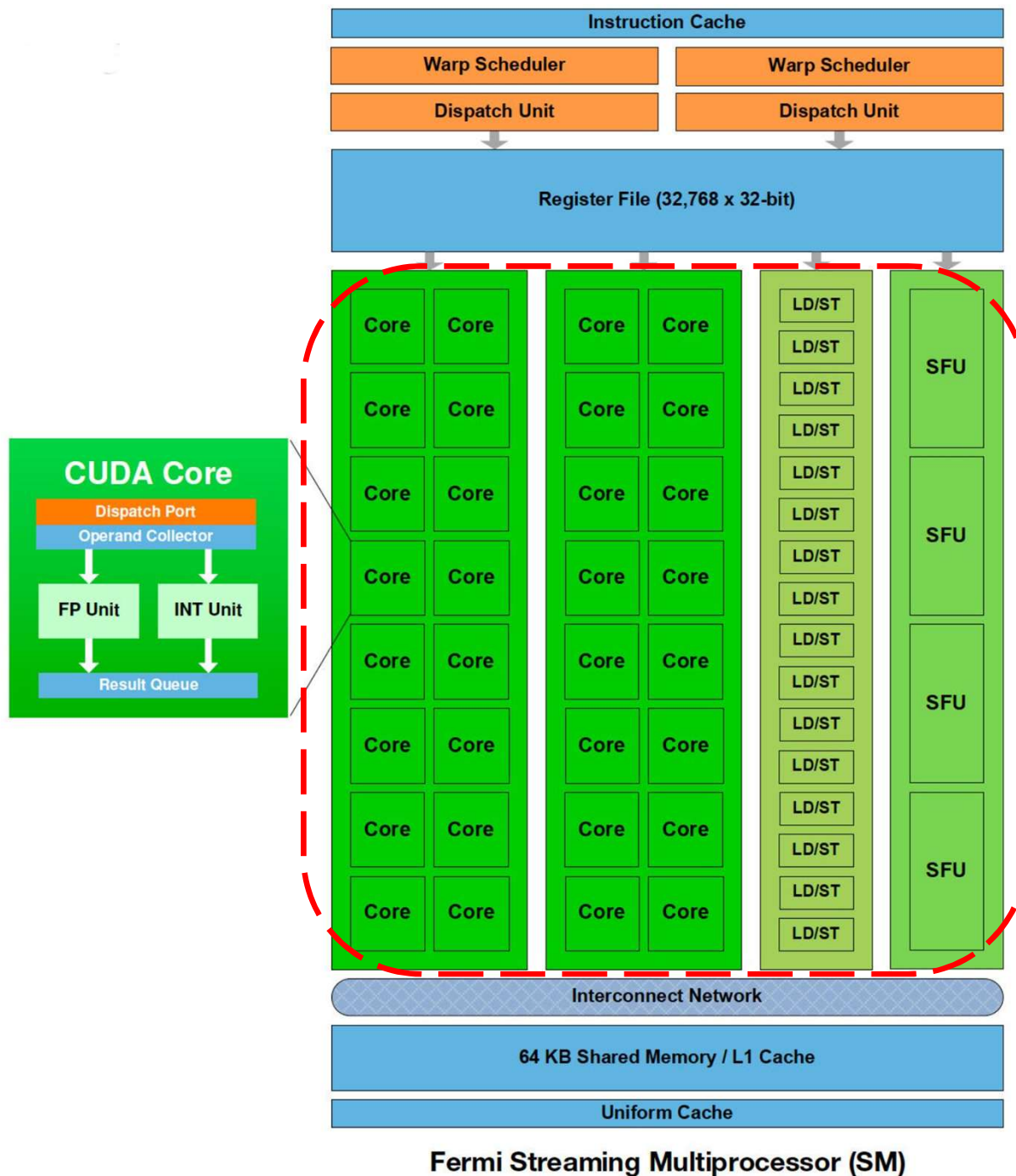
Architecture of a Fermi Streaming Multiprocessor (SM)



3. Register File (Critical Resource)

- **Register File (32K × 32-bit).**
- Each thread has its own registers, and since thousands of threads can be active, this register file must be very large.
- This is a critical performance factor:
- If registers are insufficient → spilling to memory → performance drops

Architecture of a Fermi Streaming Multiprocessor (SM)



4. Execution Units

(1) CUDA Cores

- These are the main execution units.
- Each core can perform:
 - Floating-point operations (FP Unit)
 - Integer operations (INT Unit)
- They execute instructions in a SIMD-like fashion across threads in a warp.

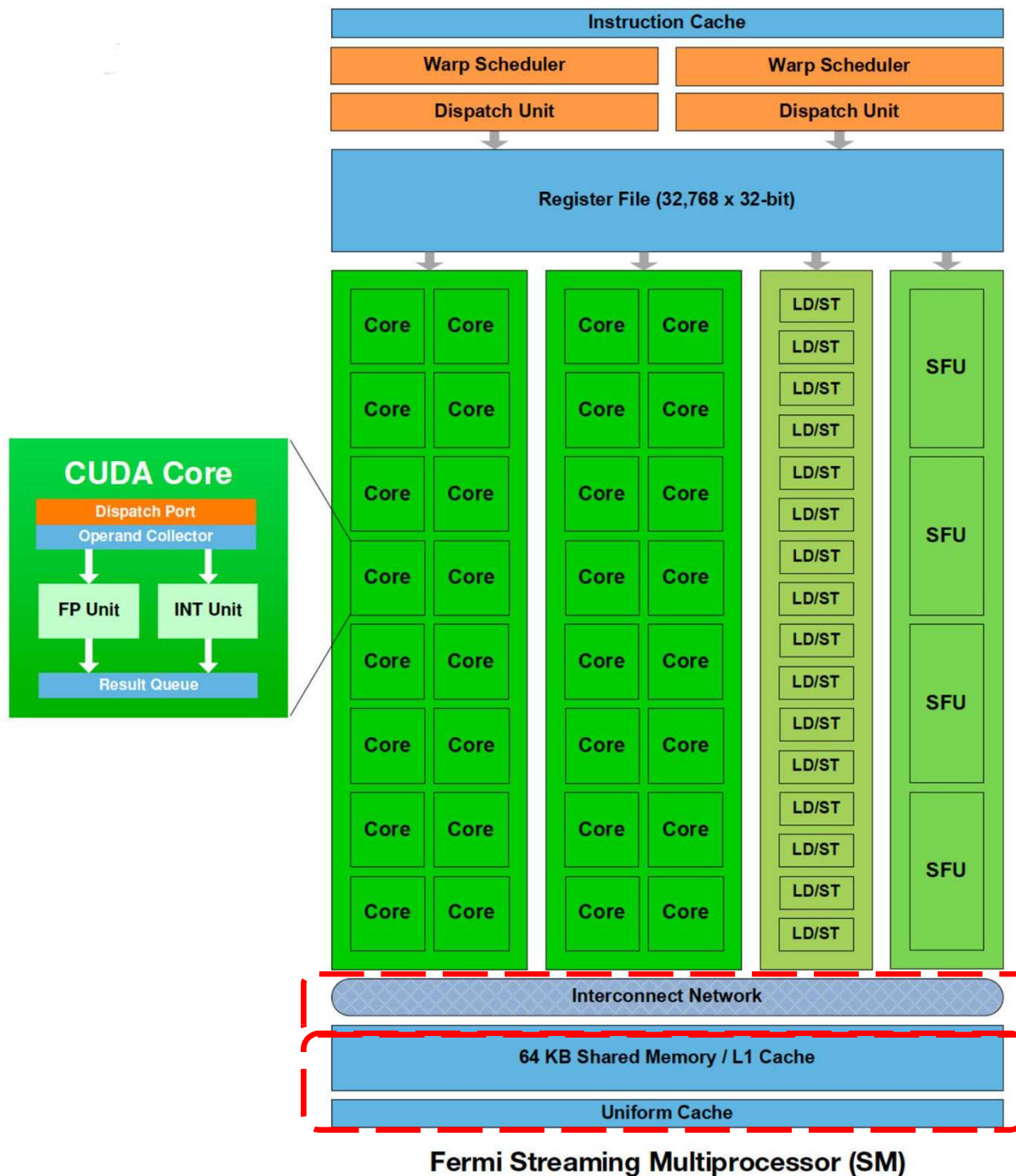
(2) Load/Store Units (LD/ST)

- These handle memory operations.
 - Global memory access
 - Data movement between memory and registers
- These operations are high latency, which is why GPU relies on warp switching.

(3) Special Function Units (SFU)

- These handle complex math functions such as: sin, cos, exp

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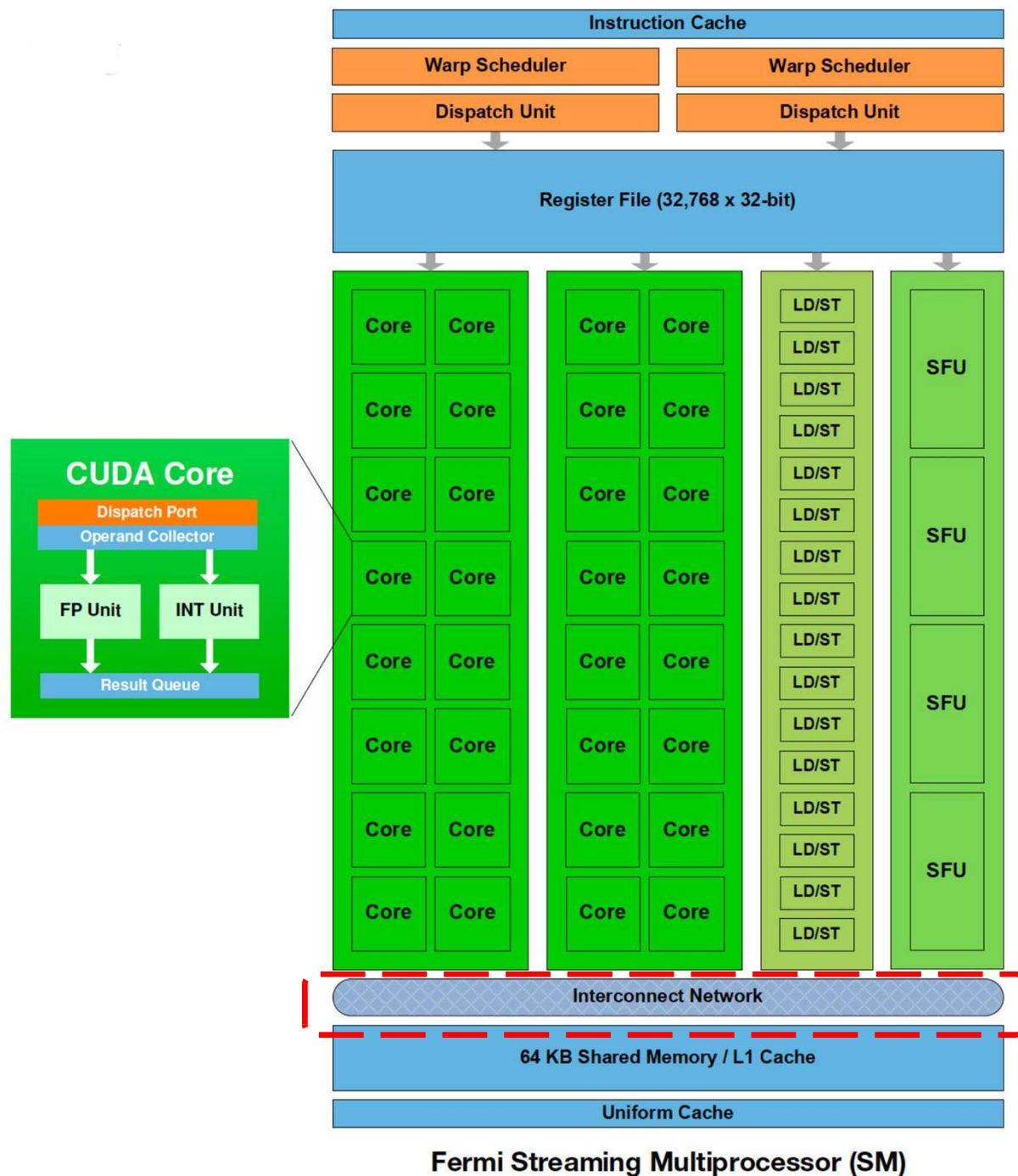
5. Interconnect Network

- This connects all internal components:
 - Register file
 - Execution units
 - Memory system

6. Shared Memory / L1 Cache

- 64KB Shared Memory / L1 Cache.**
- This is a fast, low-latency memory:
 - Shared memory → explicitly managed by programmer
 - L1 cache → automatically managed
- It is heavily used for:
 - Data reuse
 - Communication between threads in a block

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7. Key Insight

- The GPU hides latency by switching between warps.
- If one warp stalls due to memory access, another warp is immediately scheduled.
- This is fundamentally different from CPUs, which rely more on caching and speculation.